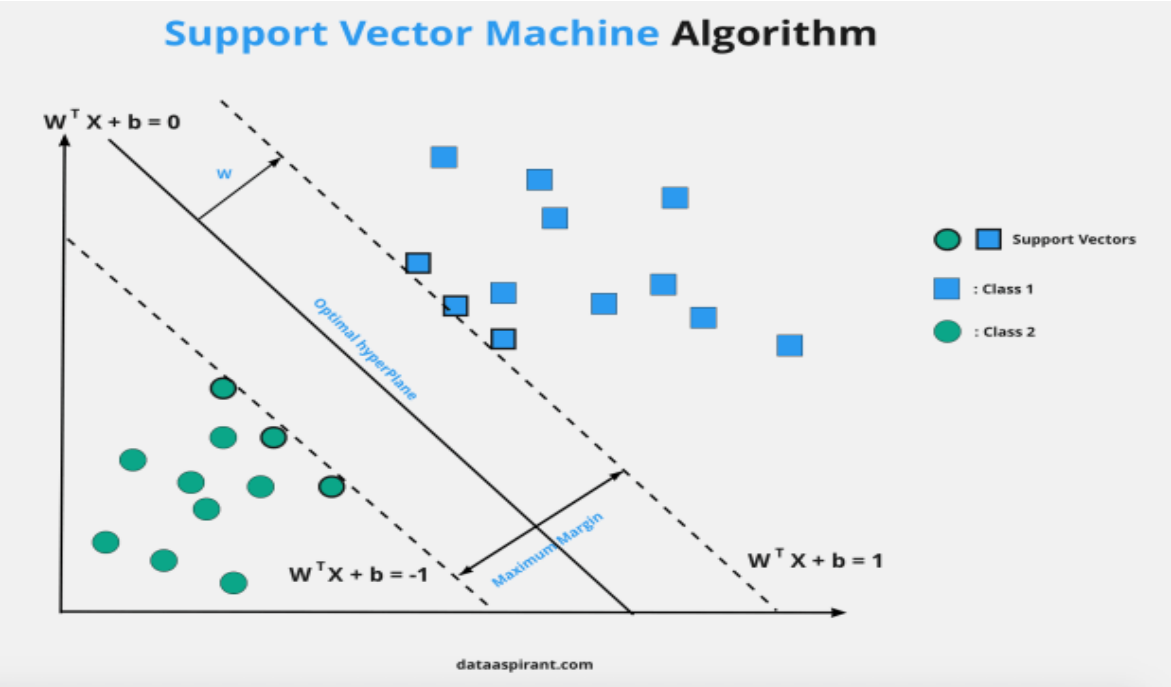
Support Vector Machine Algorithm

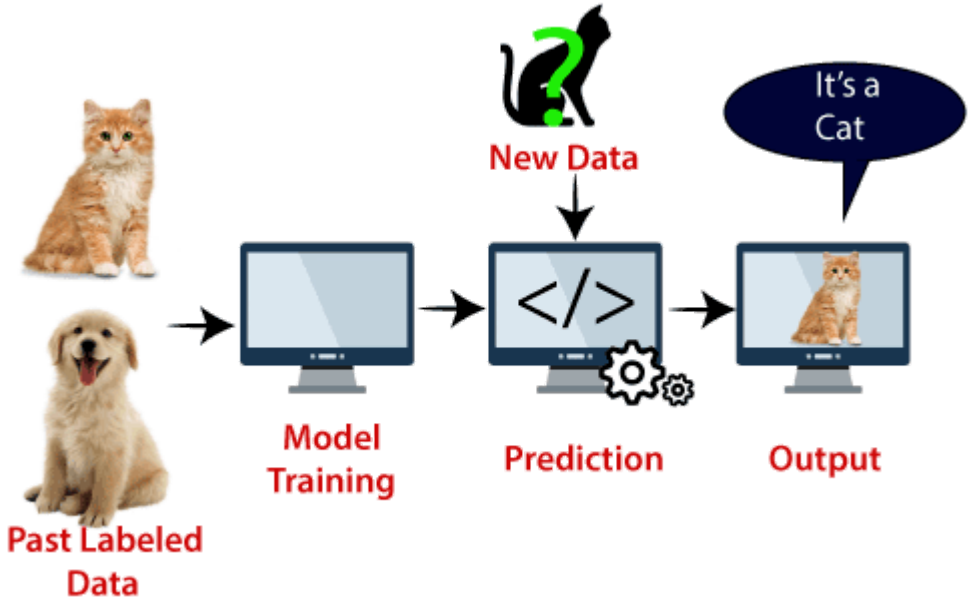
Support Vector Machine or SVM is one of the most popular Supervised Learning algorithms, which is used for Classification as well as Regression problems. However, primarily, it is used for Classification problems in Machine Learning.

The goal of the SVM algorithm is to create the best line or decision boundary that can segregate n-dimensional space into classes so that we can easily put the new data point in the correct category in the future. This best decision boundary is called a hyperplane.

SVM chooses the extreme points/vectors that help in creating the hyperplane. These extreme cases are called as support vectors, and hence algorithm is termed as Support Vector Machine. Consider the below diagram in which there are two different categories that are classified using a decision boundary or hyperplane:



Example: SVM can be understood with the example that we have used in the KNN classifier. Suppose we see a strange cat that also has some features of dogs, so if we want a model that can accurately identify whether it is a cat or dog, so such a model can be created by using the SVM algorithm. We will first train our model with lots of images of cats and dogs so that it can learn about different features of cats and dogs, and then we test it with this strange creature. So as support vector creates a decision boundary between these two data (cat and dog) and choose extreme cases (support vectors), it will see the extreme case of cat and dog. On the basis of the support vectors, it will classify it as a cat. Consider the below diagram:



SVM algorithm can be used for **Face detection, image classification, text categorization,** etc.

Types of SVM

SVM can be of two types:

- **Linear SVM:** Linear SVM is used for linearly separable data, which means if a dataset can be classified into two classes by using a single straight line, then such data is termed as linearly separable data, and classifier is used called as Linear SVM classifier.
- **Non-linear SVM:** Non-Linear SVM is used for non-linearly separated data, which means if a dataset cannot be classified by using a straight line, then such data is termed as non-linear data and classifier used is called as Non-linear SVM classifier.

Hyperplane and Support Vectors in the SVM algorithm:

Hyperplane: There can be multiple lines/decision boundaries to segregate the classes in n-dimensional space, but we need to find out the best decision boundary that helps to classify the data points. This best boundary is known as the hyperplane of SVM.

The dimensions of the hyperplane depend on the features present in the dataset, which means if there are 2 features (as shown in image), then hyperplane will be a straight line. And if there are 3 features, then hyperplane will be a 2-dimension plane.

We always create a hyperplane that has a maximum margin, which means the maximum distance between the data points.

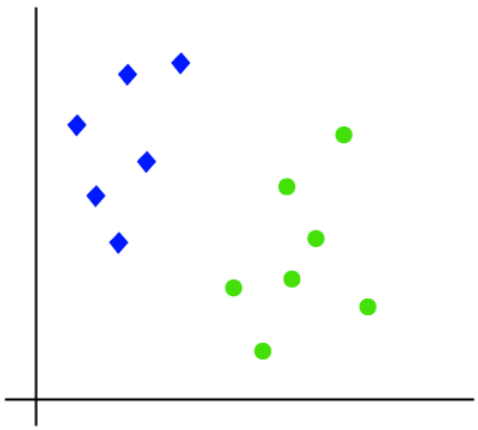
Support Vectors:

The data points or vectors that are the closest to the hyperplane and which affect the position of the hyperplane are termed as Support Vector. Since these vectors support the hyperplane, hence called a Support vector.

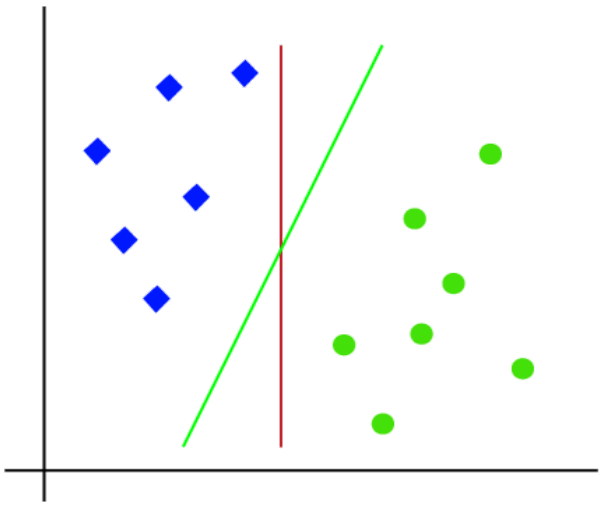
How does SVM works?

Linear SVM:

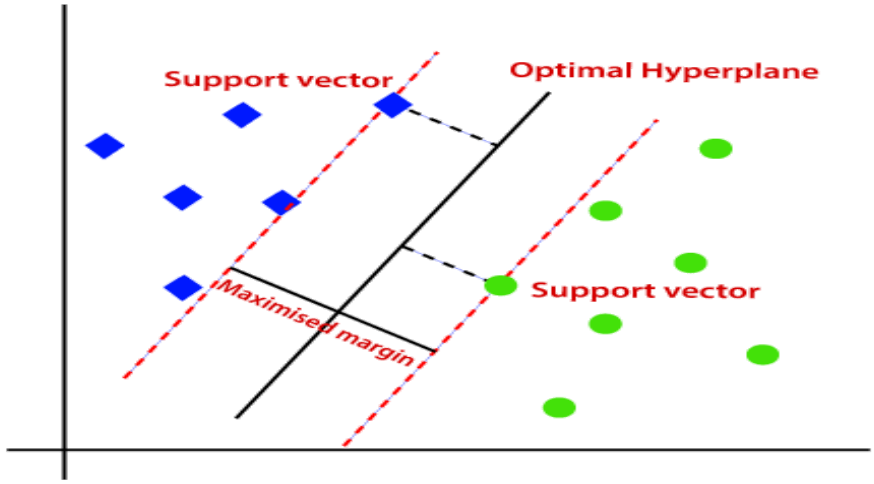
The working of the SVM algorithm can be understood by using an example. Suppose we have a dataset that has two tags (green and blue), and the dataset has two features x1 and x2. We want a classifier that can classify the pair(x1, x2) of coordinates in either green or blue. Consider the below image:



So as it is 2-d space so by just using a straight line, we can easily separate these two classes. But there can be multiple lines that can separate these classes. Consider the below image:

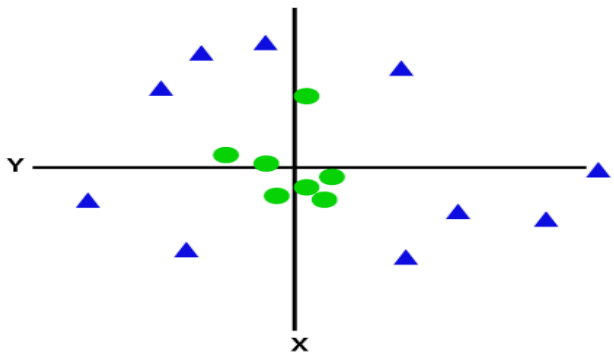


Hence, the SVM algorithm helps to find the best line or decision boundary; this best boundary or region is called as a **hyperplane**. SVM algorithm finds the closest point of the lines from both the classes. These points are called support vectors. The distance between the vectors and the hyperplane is called as **margin**. And the goal of SVM is to maximize this margin. The **hyperplane** with maximum margin is called the **optimal hyperplane**.



Non-Linear SVM:

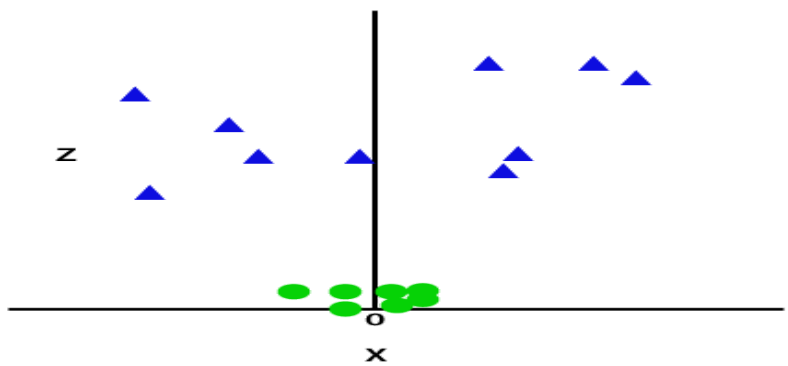
If data is linearly arranged, then we can separate it by using a straight line, but for non-linear data, we cannot draw a single straight line. Consider the below image:



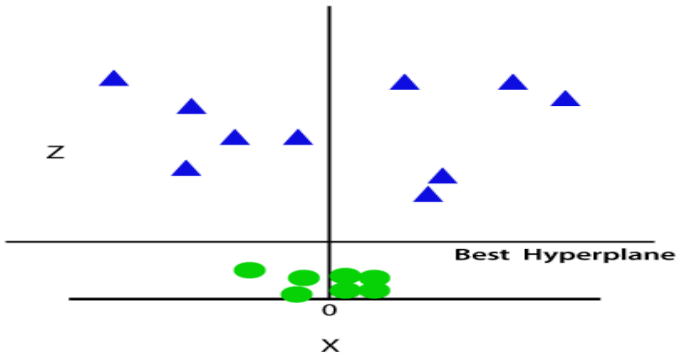
So to separate these data points, we need to add one more dimension. For linear data, we have used two dimensions x and y, so for non-linear data, we will add a third dimension z. It can be calculated as:

$z=x^2+y^2$

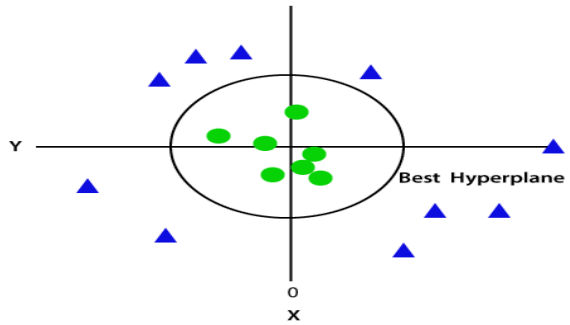
By adding the third dimension, the sample space will become as below image:



So now, SVM will divide the datasets into classes in the following way. Consider the below image:



Since we are in 3-d Space, hence it is looking like a plane parallel to the x-axis. If we convert it in 2d space with z=1, then it will become as:



Hence we get a circumference of radius 1 in case of non-linear data

Linear *Inseparability*

Before we move on to the concepts of Soft Margin and Kernel trick, let us establish the need of them. Suppose we have some data and it can be depicted as following in the 2D space:

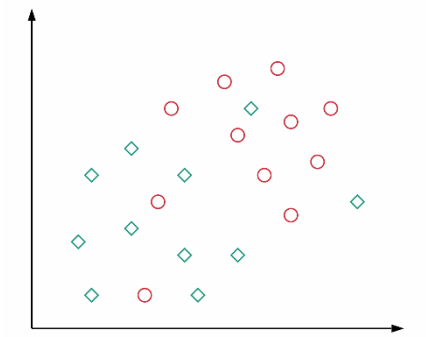


Figure 1: Data representation where the two classes are not linearly separable

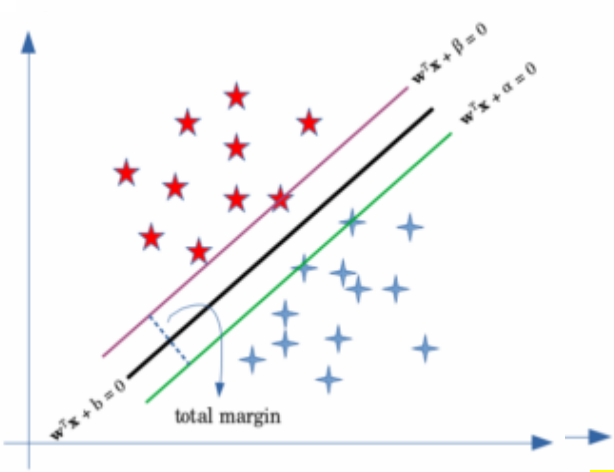
From the figure, it is evident that there's no specific linear decision boundary that can perfectly separate the data, i.e. *the data is linearly inseparable*. We can have a similar situation in higher-dimensional representations as well. This can be attributed to the fact that usually,

the features we derive from the data don't contain *sufficient information* so that we can clearly separate the two classes. This is usually the case in many real-world applications

Hard Margin vs. Soft Margin

The difference between a hard margin and a soft margin in SVMs lies in the separability of the data. If our data is linearly separable, we go for a hard margin. However, if this is not the case, it won't be feasible to do that. In the presence of the data points that make it impossible to find a linear classifier, we would have to be more lenient and let some of the data points be misclassified. In this case, a soft margin SVM is appropriate.

Sometimes, the data is linearly separable, but the margin is so small that the model becomes prone to overfitting or being too sensitive to outliers. Also, in this case, we can opt for a larger margin by using soft margin SVM in order to help the model generalize better.



SVM with a Hard Margin SVM with a Soft Margin

SVM Applications

SVMs are utilized in applications like

- **Handwriting recognition**,
- Intrusion detection,
- **Face detection**,
- **Email classification**,
- Gene classification.

That's why we prefer SVMs in various **machine learning applications**. Also, it can handle both **classification and regression** on linear and non-linear data.

Another reason we use SVMs is that they help us to find complex relationships among the provided dataset without you being involved in plenty of transformations on your own.

It is a great algorithm to choose when you are working with smaller datasets that have tens to many thousands of features. They typically find more accurate results when put next to **other algorithms** due to their ability to handle small, complex datasets.

SVM Kernel Functions

SVM algorithms use a group of mathematical functions that are known as kernels. The function of a kernel is to require data as input and transform it into the desired form.

Different SVM algorithms use differing kinds of kernel functions. These functions are of different kinds—for instance, linear, nonlinear, polynomial, radial basis function (RBF), and sigmoid.

The most preferred kind of kernel function is RBF. Because it's localized and has a finite response along the complete x-axis.

The kernel functions return the scalar product between two points in an exceedingly suitable feature space. Thus by defining a notion of resemblance, with a little computing cost even in the case of very high-dimensional spaces.

$$K(\overline{x}) = \begin{matrix} 1 \\ 0 \end{matrix} \quad \begin{matrix} \text{if} \|\overline{x}\| \leq 1 \\ \text{otherwise} \end{matrix}$$

Popular SVM Kernel Functions

Linear Kernel

It is the most basic type of kernel, usually one dimensional in nature. It proves to be the best function when there are lots of features. The linear kernel is mostly preferred for **text classification problems** as most of these kinds of classification problems can be linearly separated.

Linear kernel functions are **faster** than other functions.

Linear Kernel Formula

$$F(x, xj) = sum(x.xj)$$

Here, **x, xj** represents the data you're trying to classify.

Polynomial Kernel

It is a more generalized representation of the linear kernel. It **is not** as preferred as other kernel functions as it is **less efficient** and accurate.

Polynomial Kernel Formula

$$F(x, xj) = (x.xj+1)^d$$

Here '.' shows the **dot product** of both the values, and **d** denotes the degree.

F(x, xj) representing the **decision boundary** to separate the given classes.

Gaussian Radial Basis Function (RBF)

It is one of the most preferred and used kernel functions in svm. It is usually chosen for non-linear data. It helps to make proper separation when there is no prior knowledge of data.

Gaussian Radial Basis Formula

$$F(x, xj) = exp(-gamma * ||x - xj||^2)$$

The value of gamma varies from **0 to 1**. You have to manually provide the value of gamma in the code. The most preferred value for **gamma is 0.1**.

Sigmoid Kernel

It is mostly preferred for **neural networks**. This kernel function is similar to a two-layer perceptron model of the neural network, which works as an **activation function** for neurons.

It can be shown as,

Sigmoid Kenel Function

F(x, xj) = tanh(αxay + c)

Gaussian Kernel

It is a commonly used kernel. It is used when there is no prior knowledge of a given dataset.

Gaussian Kernel Formula

$$k(x, y) = \exp\left(-\frac{||x - y||^2}{2\sigma^2}\right)$$

Bessel function kernel

It is mainly used for removing the cross term in mathematical functions.

Bassel Kernel Formula

$$k(x, y) = \frac{J_{v+1}(\sigma||x - y||)}{||x - y||^{-n(v+1)}}$$

Here J is the Bessel function.

ANOVA kernel

It is also known as a radial basis function kernel. It usually performs well in multidimensional regression problems.

Anova Kernel Formula

$$k(x, y) = \sum_{k=1}^{n} \exp(-\sigma(x^k - y^k)^2)^d$$